

Networks Decide for Us

Phase Transitions and Universal Scaling in Homophily-Imputed Adoption Cascades

Aníbal Olivera Morales

Centro de Investigación en Complejidad Social (CICS)
Universidad del Desarrollo, Santiago, Chile

with Jorge Fábrega Lacoa & Thomas W. Valente

Critical Phenomena in Networks (CPIN)

Why Networks Matter for Adoption

- Collective behaviors (innovations, social movements, beliefs) spread through networks.
- Classical models: homogeneous populations, synthetic topologies.
- Reality: people cluster by similarity (**homophily**).
- **Question:** Does empirical network structure act as an autonomous causal force?
- **Hypothesis:** Homophilic clustering governs both the *volume* and *dynamics* of collective change.

From Physics to Society: The Mapping

Ferromagnetic System (Ising)

- Spins: align or anti-align
- Temperature T : disorder
- Field H : external bias
- Order parameter: Magnetization M

Social Adoption

- Agents: adopt or abstain
- **MSP** (Max Social Proximity): social flexibility
- **IUL** (Intrinsic Utility): innovation attractiveness
- Order parameter: Adoption Φ

Low MSP = rigid clusters (frozen)		High MSP = flexible (melted)
-----------------------------------	--	------------------------------

1 Network Imputation

- Homophily parameters from GSS core discussion networks (Smith et al., 2014)
- ERGM simulation onto ATP 2014 respondents ($N = 1000$)

2 Agent-Based Model (netdiffuseR)

- Hybrid: Rational adoption (q_i) + *Socialinfluence(selectivebysimilarity)*
- 4 million simulations sweeping IUL $\times$ MSP $\times$ threshold parameters

3 Analysis

- Generalized Additive Models (GAM): measure structural premium
- Phase transition analysis: extract critical exponents ()

Result 1: The Structural Premium

- **Finding:** Randomizing network structure $\Rightarrow$ 57% drop in adoption.
- **Interpretation:** Homophilic clustering acts as a structural *catalyst*, not just a passive conduit.
- Local echo chambers allow social reinforcement, even when population is averse to innovation.

$$\text{structure} = -0.850 \quad (p < 0.001)$$

*Volume: Networks amplify diffusion. But do they also shape **how** it happens?*

Result 2: Empirical Criticality

In empirical networks:

- **Mean-Field universality:** $\gamma \approx 0.977 \approx 1$
- Susceptibility diverges as power law: $\chi \sim |MSP - MSP_c|^{-\gamma}$
- Critical Slowing Down observed
- Continuous phase transition (2nd order)

In randomized networks:

- **No universality:** $\gamma \approx 4.0$ (unphysical)
- Power-law breaks down
- Abrupt regime shifts
- Discontinuous transition (1st order)

$$\text{Separation ratio} = \frac{\gamma_{\text{random}}}{\gamma_{\text{empirical}}} \approx 4.3 \times$$

The Paradox: Why Mean-Field in Clustered Networks?

The Puzzle

Mean-Field universality ($\gamma = 1$) typically requires well-mixed systems.
But we observe it in highly clustered, homophilic networks.

The Resolution

Homophilic clusters act as quasi-isolated *subsystems*.
At criticality ($\text{MSP} \approx \text{MSP}_c$), weak inter-cluster bridges suddenly activate.
This global synchronization is indistinguishable from mean-field behavior.

Homophily stabilizes local order & enables global coherence simultaneously.

What This Means

Networks don't just affect volume—they govern continuity.

Predictability Homophilic topologies produce legible cascades (scaling laws).

Volatility Random topologies produce sudden regime shifts (no warning).

Scale-Free At the critical boundary, adoption sizes follow power laws—fractals of social change.

Sociological Viscosity Empirical homophily acts as a *stabilizing* force that prevents chaos.

Current Constraints:

- MSP as global constant (should be heterogeneous by group/individual)
- Static topology (no co-evolution)
- $N = 1000$ (finite-size scaling analysis pending)

Natural Extensions:

- Heterogeneous MSP distributions by demographic groups
- Adaptive networks (rewiring as behavior spreads)
- Multi-layer topology (overlap across social dimensions)

Networks Decide for Us

Society's structural layout acts as an active mechanism that:

- ① Amplifies collective outcomes (structural premium)
- ② Determines whether change is continuous or catastrophic (criticality)
- ③ Obeys the same universal laws as physical phase transitions

The hidden assumption of homogeneous populations was masking a deeper truth: our networks are not neutral stages for social action—they are the fundamental choreographers of collective behavior.

Thank You

Questions?

an.oliveram@udd.cl

github.com/aoliveram/Networks-Decide-for-Us